Gerard Leonard & Cathal Flood (C22312211)

Cover page

Project Proposal Document

Contents list

1.0 Overview

With an industrial kitchen / restaurant, it is important that the correct systems are in place to allow the business to be profitable. On the other side of this, this is not a cheap endeavour. There needs to be a balance struck between cost, overheads and ultimately potential profit.
This is where a suggestion of a unique IoT focused dishwasher is a perfect solution. Allowing for kitchen staff to be more efficient with the management of kitchen and restaurant materials.
For this to be successfully implemented, there are several areas which need to be addressed. My two main areas of focus were:
Project Support
Project Manufacturing

2.0 Objectives or scope

Project Manufacturing:

- Four sub focus areas.
- Component Manufacturing.
  - This covers logistical planning, cost management, and supplier evaluation.
- Quality Assurance.
  - This covers vital areas such as system integrity, performance under load, and final QA testing.
- IoT Hardware Manufacturing.
  - This covers the development and manufacture of the sensor components, and the successful integration into the final product.
- The Production aspect of manufacturing.
  - This ensures that the production is logical, modular, staff are correctly trained to be safe, efficient and reliable.

Project Support:

- Four sub focus areas.
- Pre-Install support
  - Covers site assessment, System customisation, and preparation of the space
- User Training
  - Comprises of hands-on training, Online resource creation, and ensuring there is support available if needed.
- Post install support
  - Covers remote monitoring of systems, troubleshooting and resolving any detected faults / issues.

- Maintenance in place covering software updates, as well as preventative / scheduled maintenance.

3.0 General approach

Managerial Approach

1. **Project Planning and Scope Definition**:
    - Goals: To manufacture and install an industrial IoT enabled dishwasher system.
    - Deliverables: software for the system has been developed.
    - Milestones: First unit will have been manufactured and dispatched to customers.
    - The project from start to finish will run in 6-month cycles starting with hardware manufacture, IoT integration followed by installation on-prem.
2. **Risk Management**:
    - If there are component shortages, we will have alternate suppliers in place to deal with this.
    - Progress will be reviewed monthly to ensure the project remains aligned with the initial vision / overview of the project.

3.1 Stakeholder Identification

The stakeholders we will be looking to gain funding from would mainly come from the hospitality sector. This is due to it being our main target audience. As a secondary audience, we could cater to large companies / businesses who have kitchen / canteen facilities on their premises.

3.2 Stake Holder Communication

Stakeholders are a key part of the project. They will be kept abreast of any developments and updated monthly. This is to ensure that they are happy with the direction in which the project is progressing.

4.0 Contractual aspects

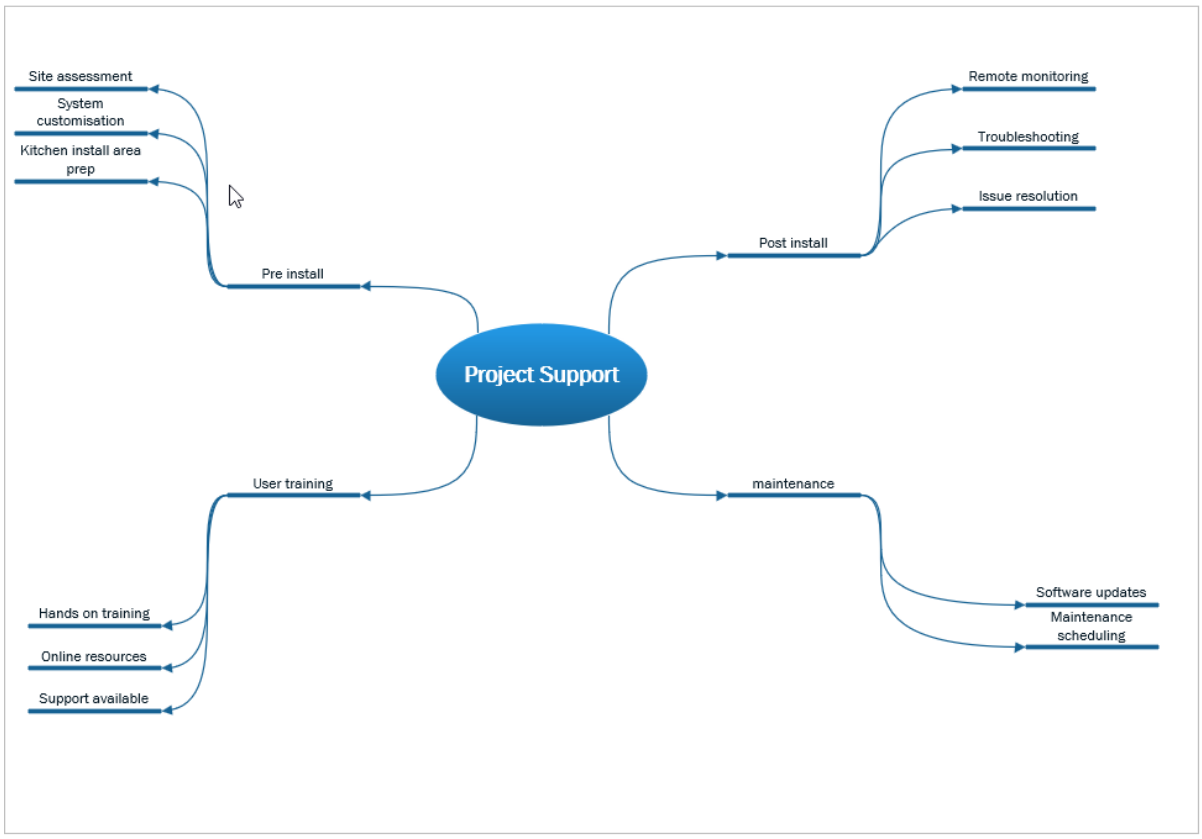
Contracts will define scope, deliverables, and timelines. Regular progress reports will ensure transparency, highlighting milestones, as well as to identify any potential risks. Technical specifications will be outlined in the planning stage with customers.

A dedicated liaison will serve as the primary contact between stakeholders. This allows for there to be seamless communication. Advisory committees will guide important decisions. Regular project reviews will assess the progress of the project. Provisioning measures for cancellation will be put in place in the case of significant delays or issues.

5.0 Project Planning

5.1 Mind map

Description



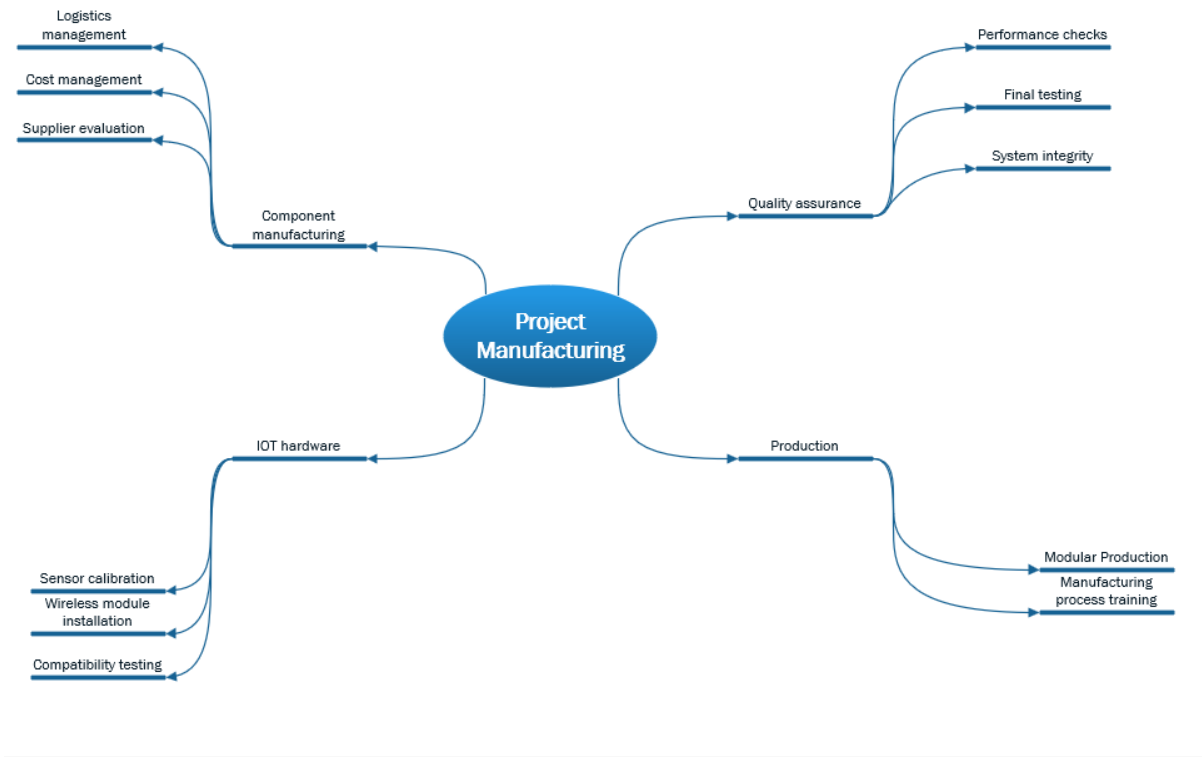


Figure 5.1.2 Project Manufacturing

5.2 Work Breakdown Structures

5.2.1 Site Assessment

Description

The Steps needed to conduct a Site Assessment are as follows:

- 1.1 Understand Requirements
- 1.2 Evaluate existing space
- 1.3 Check the layouts
- 1.4 Evaluate environment
- 1.5 Check utilities
- 1.6 Test network
- 1.7 Compliance review
- 1.8 Integration with current systems
- 1.9 Document plan

Reference to website

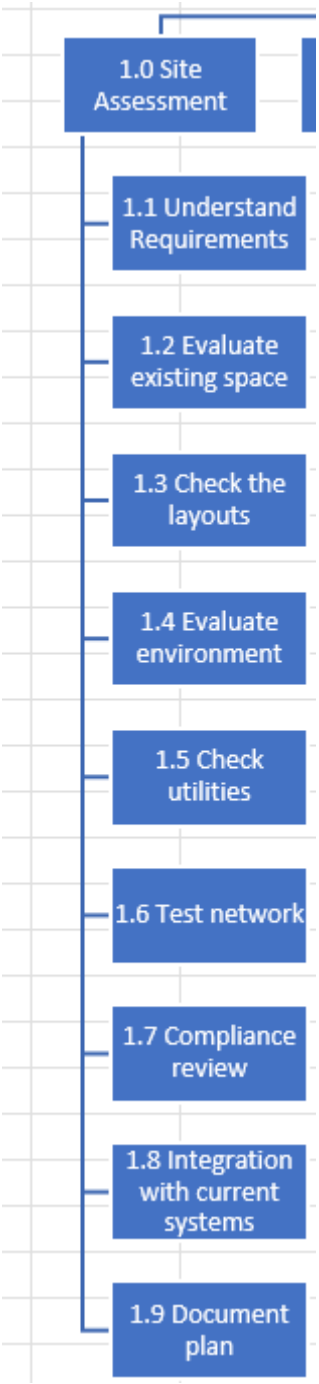


Figure 5.2.1 Work Package 1 Conduct a site assessment

5.2.2 System Customisation

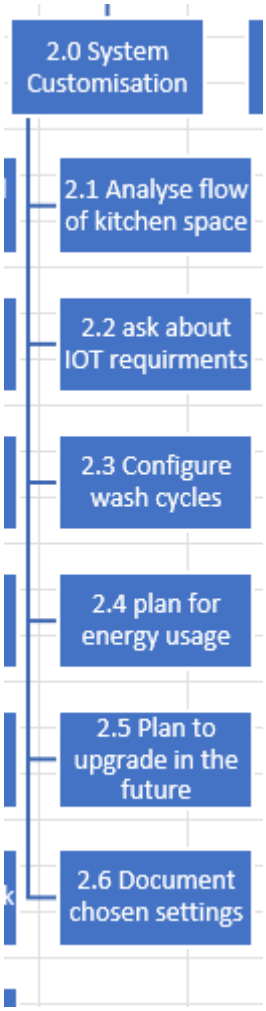


Figure 5.2.2 Work Package 2 prepare to customise the system

Description

The steps needed to customise the system are as follows:

- 2.1 Analyse flow of kitchen space
- 2.2 ask about IOT requirements
- 2.3 Configure wash cycles
- 2.4 plan for energy usage
- 2.5 Plan to upgrade in the future
- 2.6 Document chosen settings

5.2.3 Kitchen Install Prep

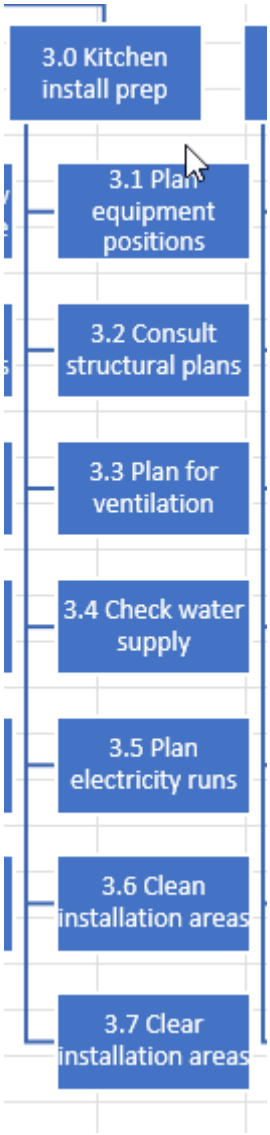


Figure 5.2.3 Work Package 3 prepare to install solution

Description

The steps needed to prep the kitchen are as follows:

- 3.1 Plan equipment positions
- 3.2 Consult structural plans
- 3.3 Plan for ventilation
- 3.4 Check water supply
- 3.5 Plan electricity runs
- 3.6 Clean installation areas
- 3.7 Clear installation areas

5.2.4 Online Resource Creation



Figure 5.2.4 Work Package 4 Create online resources

The steps needed to create online resources are as follows:

- 4.1 Identify trainig areas
- 4.2 develop training strucure
- 4.3 create videos
- 4.4 provide user manual
- 4.5 provide troubleshooting steps
- 4.6 use online platform
- 4.7 mobile friendly content

5.2.5 Hands on Training

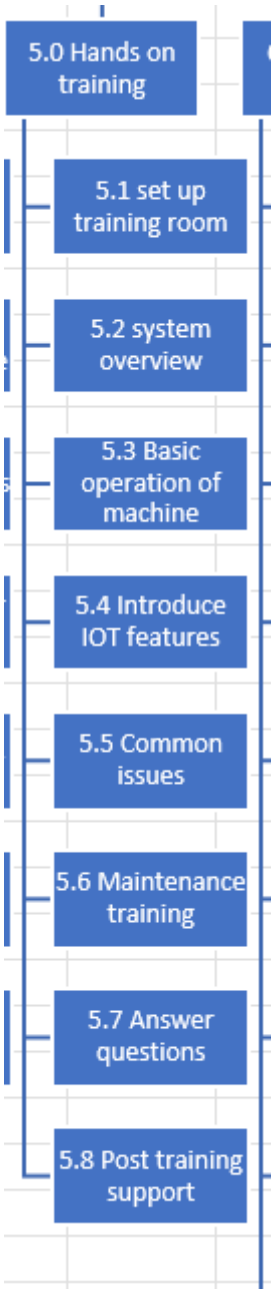


Figure 5.2.5 Work Package 5 Create hands on Training Solution

The steps needed to create a hands-on training solution are as follows:

- 5.1 set up training room
- 5.2 system overview
- 5.3 Basic operation of machine
- 5.4 Introduce IOT features
- 5.5 Common issues

- 5.6 Maintenance training
- 5.7 Answer questions
- 5.8 Post training support

5.2.6 Software Updates

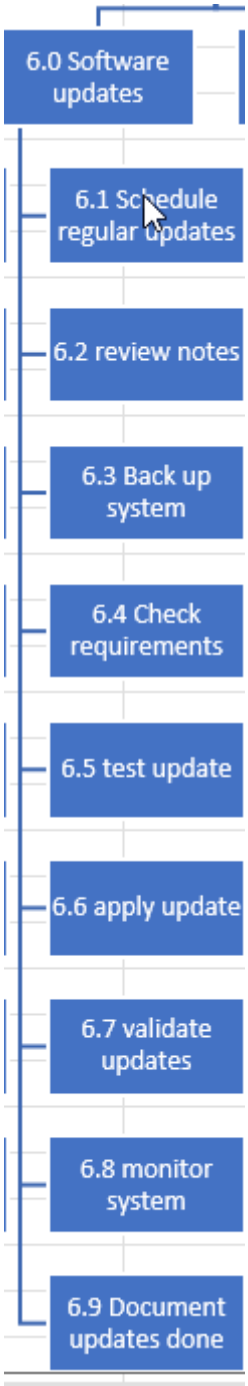


Figure 5.2.6 Work Package 6 managing software updates

The steps needed to manage software updates are as follows:

- 6.1 Schedule regular updates
- 6.2 review notes
- 6.3 Back up system
- 6.4 Check requirements
- 6.5 test update
- 6.6 apply update
- 6.7 validate updates
- 6.8 monitor system
- 6.9 Document updates done

5.2.7 Maintenance Scheduling

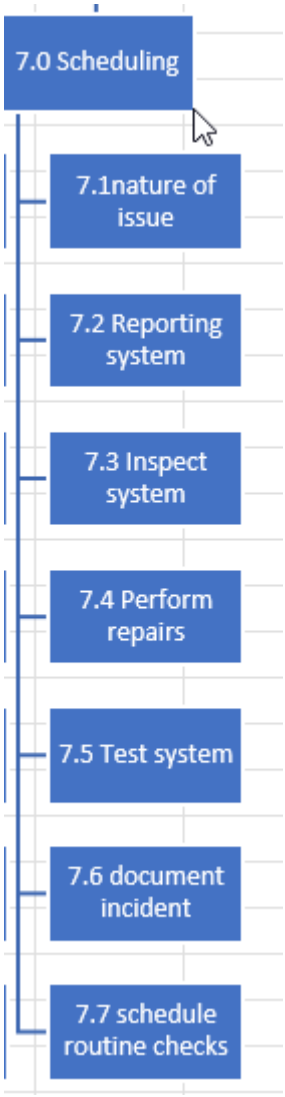


Figure 5.2.7 Work Package 7 Maintenance Scheduling

The steps needed to correctly schedule maintenance windows are:

- 7.1 Determine the nature of issue
- 7.2 Reporting system
- 7.3 Inspect system
- 7.4 Perform repairs
- 7.5 Test system
- 7.6 document incident
- 7.7 schedule routine checks

5.2.8 Logistics Management

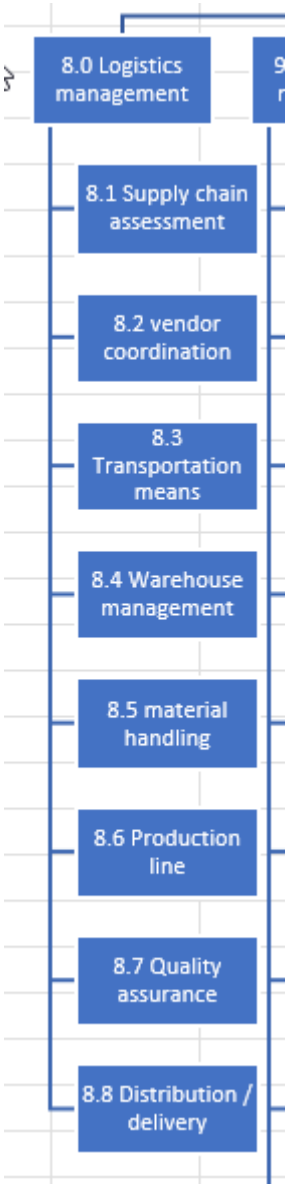


Figure 5.2.8 Work Package 8 Logistics Management

The steps needed to correctly implement logistics management are:

- 8.1 Supply chain assessment
- 8.2 vendor coordination
- 8.3 Transportation means
- 8.4 Warehouse management
- 8.5 material handling
- 8.6 Production line
- 8.7 Quality assurance
- 8.8 Distribution / delivery

5.2.9 Hardware Manufacturing

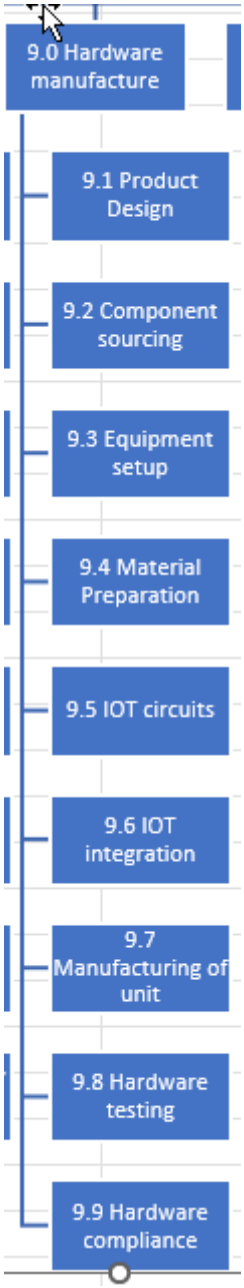


Figure 5.2.9 Work Package 9 Hardware Manufacturing

The steps needed to correctly implement hardware manufacturing are:

- 9.1 Product Design
- 9.2 Component sourcing
- 9.3 Equipment setup
- 9.4 Material Preparation
- 9.5 IOT circuits
- 9.6 IOT integration
- 9.7 Manufacturing of unit
- 9.8 Hardware testing
- 9.9 Hardware compliance

5.2.10 Cost Management

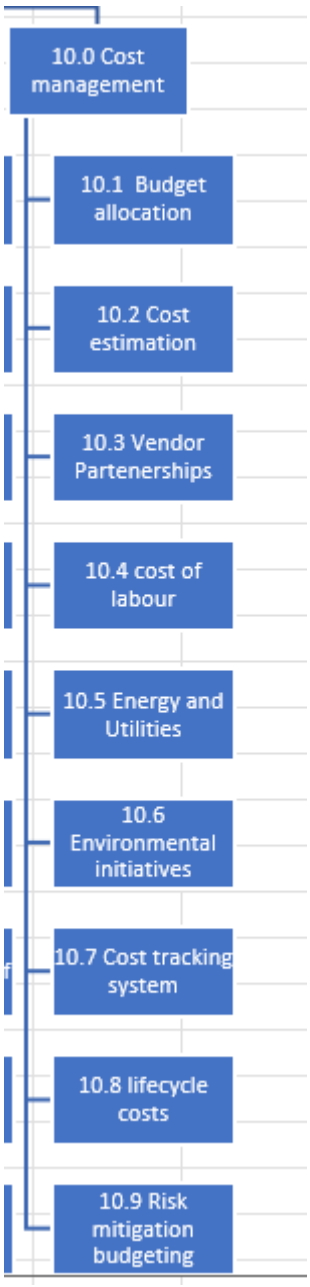


Figure 5.2.10 Work Package 10

The steps needed to correctly implement effective cost management are:

- 10.1 Budget allocation
- 10.2 Cost estimation
- 10.3 Vendor Partnerships
- 10.4 cost of labour
- 10.5 Energy and Utilities
- 10.6 Environmental initiatives
- 10.7 Cost tracking system
- 10.8 lifecycle costs

- 10.9 Risk mitigation budgeting

5.3 Action Plans

Action Plan for Work Package 1.0: Site Assessment					
Deliverables on completion of Work Package: A plan for upgrade will be completed					
Measures of Accomplishments / Milestones Site is surveyed, and utilities are checked.					
Key constraints and Assumptions There is already a non IOT solution in place					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
1.0	Site assessment				
1.1	Understand Requirements			3 days	IOT solutions architect, Project manager
1.2	Evaluate existing space		1.1	2 days	Site engineer
1.3	Check the layout		1.2	1 day	Site engineer
1.4	Evaluate environment		1.3	2 days	Design engineer

1.5	Check utilities	Tool kit	1.4	1 day	Electrician, Plumber
1.6	Test Connectivity	Laptop	1.5	2 days	IT technician
1.7	Compliance Review		1.6	2 days	Compliance officer
1.8	Integration with existing systems		1.7	3 days	Systems engineer
1.9	Document plan		1.8	2 days	Project Manager

Action Plan for Work Package 2.0: System Customisation					
Deliverables on completion of Work Package: System will be planned to the need of customer.					
Measures of Accomplishments / Milestones Project planning after visit has taken place.					
Key constraints and Assumptions Every kitchen will have different requirements and layout.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
2.0	System Customisation				
2.1	Analyse flow of kitchen space			2 days	Site engineer
2.2	Ask about IOT requirements		2.1	1 day	IOT specialist
2.3	Configure Wash cycles		2.2	2 days	Software engineer
2.4	Plan for energy usage		2.3	2 days	Energy engineer
2.5	Future planning		2.4	3 days	Project manager
2.6	Document chosen settings		2.5	1 day	Kitchen manager

Action Plan for Work Package 3.0: Kitchen install prep					
Deliverables on completion of Work Package: The area will be ready for the installation to take place					
Measures of Accomplishments / Milestones Utilities have been checked. Installation area is clear					
Key constraints and Assumptions Proper precautions have been taken without disrupting existing plans.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
3.0	Kitchen install prep				
3.1	Plan equipment positions			2 days	Design engineer
3.2	Consult structural plans		3.1	1 day	Structural engineer
3.3	Plan for ventilation		3.2	1 day	HVAC engineer
3.4	Check water supply		3.3	1 day	Plumber

3.5	Plan electricity runs		3.4	2 days	Electrician
3.6	Clean installation areas		3.5	1 day	Kitchen cleaner
3.7	Clear installation areas		3.6	1 day	Maintenance engineer

Action Plan for Work Package 4.0: Online Resources					
Deliverables on completion of Work Package: Online training resources are available.					
Measures of Accomplishments / Milestones Training structure in place. Online platform is available.					
Key constraints and Assumptions The material must be mobile friendly to facilitate learning at work.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
4.0	Online Resources				
4.1	Identify training areas			2 days	Training Manager
4.2	Develop training structure		4.1	4 days	Training manager
4.3	Create video content		4.2	7 days	IOT dishwasher expert
4.4	Develop user manual		4.3	4 days	Product specialist
4.5	Provide troubleshooting steps		4.4	2 days	Support engineer
4.6	Use online platform		4.5	3 days	Learning management system admin
4.7	Mobile friendly content		4.6	2 days	UX designer

Action Plan for Work Package 5.0: Hands on Training					
Deliverables on completion of Work Package: Staff have been trained in machine systems					
Measures of Accomplishments / Milestones Common issues have been discussed					
Key constraints and Assumptions All kitchen and management staff must learn how systems operate.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
5.0	Hands on training				
5.1	Set up training room			1 day	Training coordinator

5.2	System overview		5.1	½ a day	Product specialist
5.3	Basic operation of machine		5.2	½ a day	Product specialist
5.4	Introduce IOT features		5.3	½ a day	IOT specialist
5.5	Common issues		5.4	1 day	Support engineer
5.6	Maintenance Training		5.5	2 days	Support engineer
5.7	Answer questions		5.6	½ a day	Product specialist
5.8	Post training support		5.7	ongoing	Support engineer

Action Plan for Work Package 6.0: Software Updates					
Deliverables on completion of Work Package: Software updates for the system will have been performed					
Measures of Accomplishments / Milestones Current configuration has been backed up.					
Key constraints and Assumptions The machine must always have internet connectivity.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
6.0	Software Updates				
6.1	Schedule regular updates			1 day	IT admin
6.2	Review notes		6.1	1 day	IOT specialist
6.3	Back-up system		6.2	½ a day	Support engineer
6.4	Check update Requirements		6.3	½ a day	Software engineer
6.5	Test the update		6.4	2 days	IOT specialist
6.6	Apply the update		6.5	1 day	Software engineer
6.7	Validate update		6.6	½ a day	IOT specialist
6.8	Monitor system status		6.7	2 days	Support engineer
6.9	Document updates performed		6.8	½ a day	IT admin

Action Plan for Work Package 7.0: Maintenance Scheduling					
Deliverables on completion of Work Package: Routine maintenance will be scheduled.					
Measures of Accomplishments / Milestones Any faults have been identified and repaired.					
Key constraints and Assumptions All incidents and fixes must be documented.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
7.0	Maintenance Scheduling				

7.1	Detect issues			1 day	IOT Specialist
7.2	Issue reporting system		7.1	1 day	IT admin
7.3	Inspect the system		7.2	1 day	Maintenance technician
7.4	Perform repairs		7.3	2 days	Maintenance technician
7.5	Test the system		7.4	1 day	Support engineer
7.6	Document incidents		7.5	1 day	Maintenance supervisor
7.7	Schedule routine checks		7.6	½ a day	Support engineer

Action Plan for Work Package 8.0: Logistics Management					
Deliverables on completion of Work Package: There must be transportation of assets available.					
Measures of Accomplishments / Milestones Warehouse management system is online. Quality assurance is in place.					
Key constraints and Assumptions There will be contracted drivers who can also deliver equipment directly to site.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
8.0	Logistics Management				
8.1	Supply Chain Assessment			3 days	Logistics manager
8.2	Vendor Coordination		8.1	2 days	Procurement specialist
8.3	Means of Transport		8.2	1 day	Transport coordinator
8.4	Warehouse Management		8.3	2 days	Warehouse manager
8.5	Material Training		8.4	4 days	Warehouse supervisor
8.6	Production line Management		8.5	2 days	Logistics manager
8.7	Quality assurance		8.6	3 days	Warehouse manager
8.8	Distribution / delivery		8.7	2 days	Contracted drivers

Action Plan for Work Package 9.0:					
Deliverables on completion of Work Package: Hardware will have been designed and manufactured					
Measures of Accomplishments / Milestones IOT components created.					
Key constraints and Assumptions Each solution will be manufactured to the needs of the customer.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
9.0	Hardware Manufacture				

9.1	Solution design			4 days	Design engineer
9.2	Component Sourcing		9.1	6 days	Supply chain manager
9.3	Equipment Setup		9.2	3 days	Production manager
9.4	Material Preparation		9.3	2 days	Production manager
9.5	IOT circuit design		9.4	4 days	Electrical Engineer
9.6	IOT sensor integration		9.5	3 days	Electrical Engineer
9.7	Manufacturing of machine		9.6	8 days	Assembly worker
9.8	Hardware testing		9.7	4 days	Hardware Engineer
9.9	Hardware compliance checks		9.8	3 days	Compliance officer

Action Plan for Work Package 10.0: Cost Management					
Deliverables on completion of Work Package: Budget will have been completed.					
Measures of Accomplishments / Milestones Labour cost will have been established. Costs will be tracked.					
Key constraints and Assumptions Risk must be budgeted for. If there is a component shortage / dip in the hospitality industry.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
10.0	Cost Management				
10.1	Budget Allocation			3 days	Accountant
10.2	Cost estimation per area		10.1	2 days	Project manager
10.3	Vendor Partnerships		10.2	4 days	Procurement manager
10.4	Cost of labour		10.3	2 days	Accountant
10.5	Energy and Utilities		10.4	2 days	Operations manager
10.6	Environmental initiatives		10.5	2 days	Operations manager
10.7	Cost tracking system		10.6	4 days	IT admin
10.8	Lifecycle cost		10.7	3 days	Project Manager
10.9	Risk mitigation budgeting		10.8	4 days	Risk manager

Repeat this for each work package.

6.0 Project Scheduling

	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Ad
1		1 day?	Mon 25/11/24	Mon 25/11/24			
2							
3	**Site Assessment**	**18 days?**	**Mon 25/11/24**	**Wed 18/12/24**			
4	Understand Requirements	3 days	Mon 25/11/24	Wed 27/11/24		IOT solutions Ar	
5	Evaluate existing space	2 days	Thu 28/11/24	Fri 29/11/24	4	Site engineer	
6	Check the layout	1 day	Mon 02/12/24	Mon 02/12/24	5	Site engineer	
7	Evaluate environment	2 days	Tue 03/12/24	Wed 04/12/24	6	Design Engineer	
8	Check utilities	1 day	Thu 05/12/24	Thu 05/12/24	7	Electrician,Plum	
9	Test Connectivity	2 days	Fri 06/12/24	Mon 09/12/24	8	IT technician	
10	Compliance Review	2 days	Tue 10/12/24	Wed 11/12/24	9	Compliance offi	
11	Integration with existing systems	3 days	Thu 12/12/24	Mon 16/12/24	10	Systems Engine	
12	Document plan	2 days	Tue 17/12/24	Wed 18/12/24	11	Project Manager	
13							
14	**System Customisation**	**11 days?**	**Thu 19/12/24**	**Thu 02/01/25**			
15	Analyse flow of kitchen space	2 days	Thu 19/12/24	Fri 20/12/24	12	Site engineer	
16	Ask about IOT requirements	1 day	Mon 23/12/24	Mon 23/12/24	15	IOT specialist	
17	Configure Wash cycles	2 days	Tue 24/12/24	Wed 25/12/24	16	Software engine	
18	Plan for energy usage	2 days	Thu 26/12/24	Fri 27/12/24	17	Energy engineer	
19	Future planning	3 days	Mon 30/12/24	Wed 01/01/25	18	Project Manager	
20	Document chosen settings	1 day	Thu 02/01/25	Thu 02/01/25	19	Kitchen manage	
21							
22	**Kitchen install prep**	**9 days?**	**Thu 19/12/24**	**Tue 31/12/24**	**12**		
23	Plan equipment positions	2 days	Thu 19/12/24	Fri 20/12/24	12	Design Engineer	
24	Consult structural plans	1 day	Mon 23/12/24	Mon 23/12/24	23	Structural engin	
25	Plan for ventilation	1 day	Tue 24/12/24	Tue 24/12/24	24	HVAC engineer	
26	Check water supply	1 day	Wed 25/12/24	Wed 25/12/24	25	Plumber	
27	Plan electricity runs	2 days	Thu 26/12/24	Fri 27/12/24	26	Electrician	
28	Clean installation areas	1 day	Mon 30/12/24	Mon 30/12/24	27	Cleaner	
29	Clear installation areas	1 day	Tue 31/12/24	Tue 31/12/24	28	Maintenance En	
30							
31	**Online Resources**	**24 days?**	**Thu 16/01/25**	**Tue 18/02/25**			

7.0 Allocating Resources, Equipment and Personnel

Staff Table

Resource Name	Type	Pay/hr	link
Project Manager	Work	€37	
IOT Solutions Architect	Work	€46	
Site Engineer	Work	€38	
Design Engineer	Work	€15	
Electrician	Work	€41	
Plumber	Work	€52	
IT technician	Work	€17	
Compliance Officer	Work	€23	
Systems Engineer	Work	€30	
IOT Specialist	Work	€37	
Software Engineer	Work	€19	
Energy Engineer	Work	€36	
Kitchen Manager	Work	€37	
Structural Engineer	Work	€24	
HVAC Engineer	Work	€30	
Cleaner	Work	€32	
Maintenance Engineer	Work	€15	
Training Manager	Work	€22	
IOT Dishwasher Expert	Work	€45	
Product Specialist	Work	€25	

Support Engineer	Work	€25	
LMS Admin	Work	€33	
UX Designer	Work	€30	
Training Coordinator	Work	€20	
IT admin	Work	€33	
Maintenance Technician	Work	€24	
Maintenance Supervisor	Work	€26	
Logistics manager	Work	€25	
Procurement Specialist	Work	€28	
Transport Coordinator	Work	€18	
Warehouse Manager	Work	€21	
Warehouse Supervisor	Work	€16	
Contracted Driver	Work	€30	
Supply Chain Manager	Work	€22	
Production Manager	Work	€27	
Electrical Engineer	Work	€29	
Assembly Worker	Work	€14	
Accountant	Work	€29	
Procurement Manager	Work	€31	
Operations Manager	Work	€25	
Risk Manager	Work	€40	

Cost table Year 1 – 5

Resource Name	Hours Worked /yr	Cost / yr	Cost yr 1-5
Project Manager	200	€7,400	€37,000
IOT Solutions Architect	48	€2,208	€11,040
Site Engineer	72	€2,736	€13,680
Design Engineer	120	€1,800	€9,000
Electrician	48	€1,968	€9,840
Plumber	32	€1,664	€8,320
IT technician	32	€544	€2,720
Compliance Officer	32	€736	€3,680
Systems Engineer	48	€1,440	€7,200
IOT Specialist	112	€2,128	€10,640
Software Engineer	64	€2,304	€11,520
Energy Engineer	32	€1,184	€5,920
Kitchen Manager	16	€384	€1,920
Structural Engineer	16	€480	€2,400
HVAC Engineer	48	€1,536	€7,680
Cleaner	16	€240	€1,200
Maintenance Engineer	16	€352	€1,760
Training Manager	96	€2,496	€12,480
IOT Dishwasher Expert	96	€4,320	€21,600
Product Specialist	104	€2,600	€13,000
Support Engineer	160	€4,000	€20,000
LMS Admin	48	€1,584	€7,920

UX Designer	24	€720	€3,600
Training Coordinator	16	€320	€1,600
IT admin	104	€3,432	€17,160
Maintenance Technician	40	€960	€4,800
Maintenance Supervisor	16	€416	€2,080
Logistics manager	80	€2,000	€10,000
Procurement Specialist	32	€896	€4,480
Transport Coordinator	16	€288	€1,440
Warehouse Manager	80	€1,680	€8,400
Warehouse Supervisor	64	€1,024	€5,120
Contracted Driver	32	€960	€4,800
Supply Chain Manager	88	€1,936	€9,680
Production Manager	80	€2,160	€10,800
Electrical Engineer	104	€3,016	€15,080
Assembly Worker	104	€1,456	€7,280
Accountant	72	€2,088	€10,440
Procurement Manager	56	€1,736	€8,680
Operations Manager	64	€1,600	€8,000
Risk Manager	32	€1,280	€6,400
Total Wages Cost Year 1 - 5			€341, 240

8.0 Cost analysis

Price for project = Salaries @33% + Overheads @ 33% + Margin @33 %.

If all things are equal, allowing for sensible total project costs. This would ultimately mean that outgoings would be about 682, 480 per year. The total acceptable price to charge a customer to clear our margin on top would be 1,023,720 euro. These figures are reflective of the project running twice successfully each year over a 5-year period.

This is Broken down as 341,240 euro for the wage bill for the 5 years.

Then 341,240 for any overheads such as heating, light, materials, fuel to transport resources etc.

The finally clearing a margin of 33% to give us a profit of about 341,240 in an ideal world.

8.1 Payback

Give an income of 205,000 per year with an initial investment of 137,000 with the same per year for the duration of the project. It would take until the end of year 3 give or take to break even. After 3 years, the total outgoings are 548,000 euro. Our income per year is 205,000 so it would take until most of the way through th e third year to have broken even / be in profit for this section of the project.

	Year	Cost / Outgoings	Income
Before the start	0	€137,000	0
	1	€137,000	€205,000
	2	€137,000	€205,000
	3	€137,000	€205,000
	4	€137,000	€205,000
	5	€137,000	€205,000
Total		685000	820000
Pay back	Total cost / Annual Cash flow	3.341463415	years

8.2 Net Present Value

With the net present value, aiming for a 10% hurdle rate and allowing for a 2 percent increase in inflation on an annual basis. This would bring the present value of income to 738,056.30 euro with the net present value being reflected as 601,560.30 euro.

Hurdle Rate	0.1	Companies set as the return required			
inflation	0.02	Real or projected			
Discount rate	0.12	k is Sum of the Hurdle rate plus the inflation			
				Discount Factor	Discounted cash flow
	Year	Cost / Outgoings	Income	1/(1+k)^t	
Before the start	0	-€136,496	0	1	-136,496.00
	1	-€136,496	€204,744	0.892857143	182,807.14
	2	-€136,496	€204,744	0.797193878	163,220.66
	3	-€136,496	€204,744	0.711780248	145,732.74
	4	-€136,496	€204,744	0.635518078	130,118.51
	5	-€136,496	€204,744	0.567426856	116,177.24
Total		-€818,976	1023720		
	Present Value of income				738,056.30
	Net present value (is the PV income less total outgoings)				601,560.30
	Outgoings includes wages				
	Overheads as a rule of thumb are equal to salaries				

8.3 Profitability

If the project were to run more than twice annually, with an increased hurdle rate with inflation remaining constant, there is probably room for increased profits. With the current biannual setup, it demonstrates that there is profit to be had. Scaling the solution to 3 or 4 times while maintaining similar costs / outgoings would lead to a proportional increase in profits. This would lead to continued growth for the business going forward. There would then be the excess cashflow available to branch out into new development endeavours.

9.0 Risk management plans

Design and Development Risks

- **Risk**: Incomplete or incorrect technical specifications.
    - **Mitigation**: Conduct complete requirement reviews and approvals with stakeholders.
- **Risk**: Delays in hardware or software development.
    - **Mitigation**: Use a cloud-based development process with defined outcomes as well as a buffer for if projects should run over.

2. Supply Chain Risks

- **Risk**: Component shortages or delivery delays.
    - **Mitigation**: Maintain multiple suppliers for key components and stockpile critical parts.
- **Risk**: Subcontractor performance issues.
    - **Mitigation**: Set clear expectations in contracts with penalties for delays and regular progress reviews.

3. Deployment and Operational Risks

- **Risk**: Integration issues between IoT platform and dishwasher hardware.
    - **Mitigation**: Conduct extensive testing and ensure compatibility during design phases.
- **Risk**: Data security vulnerabilities.
    - **Mitigation**: Implement strong encryption, and compliance with GDPR.

4. Customer and Market Risks

- **Risk**: Misalignment with customer needs.
    - **Mitigation**: Involve customers in early testing and incorporate feedback into the design.
- **Risk**: Delayed approvals or customer-provided resources.
    - **Mitigation**: Set realistic timelines with clear resource expectations in agreements.

10.0 Evaluation methods

1. What Will Be Measured and Monitored

- **Technical Performance**: Dishwasher efficiency (cycle time, water/energy usage), IoT connectivity will be measured.
- **Customer Satisfaction**: Usability, remote functionality, and adherence to customer requirements.
- **Project Milestones**: Adherence to timelines, budget, and conditions set out initially.

2. How Data Will Be Collected and Stored

- **Data Collection**:
    - IoT sensors will log performance metrics (e.g., water flow, temperature, energy use) in real-time.
- **Storage**:
    - Project progress and testing data will be documented in a centralized project management system.

3. How the Project Will Be Evaluated

- **Testing Phases**: Evaluate prototypes and final product through testing functional components, integration, and stress testing.
- **Stakeholder Review**: Conduct regular project reviews with stakeholders to assess alignment with goals.
- **Post-Implementation Review**: Analyse the project's success based on customer feedback, recorded data, and peer review.

10.1 Return on Investment

The return on investment formula is as follows:

Net profit from project / total investment x 100.

Net profit: 135000

Total investment/ outgoings: 685000

RoI = 135000/685000 x 100 = 19.71 % return on investment.

So, looking at the project this way, even though it takes about 3 and a half years to make back the initial investment / continued outgoings / expenses. If scaled up with multiple simultaneous projects per year, there is a larger profit margin available for the future.

10.2 Comparing projects

For this to be an unbiased and fair comparison process, there are several factors which must be considered. These are what the RoI is as a percentage, what the average payback period is, and what the net present value of the investment is. You must also compare the industry each is operating in, the sector / growth potential of the product within that business.

10.3 Control

1. **Milestone Tracking**: Use a Gantt chart to schedule and monitor completion dates and goals.
2. **Quality Assurance**: Regularly test hardware, software, and IoT features to meet industry standards.
3. **Budget Monitoring**: Track expenses vs. what was set out in the budget.

4. **Risk Management**: Update mitigation plans monthly.
5. **Change Control**: Evaluate scope, cost, and timeline impacts before approving changes.
6. **Communication**: Hold regular meetings and provide clear updates via reports.
7. **IoT Metrics**: Monitor uptime, data accuracy, and energy efficiency using real-time tools.
8. **Customer Feedback**: Gather and act on feedback during early-stage testing.

These controls ensure the project stays on track, within scope, and delivers acceptable outcomes.

Project Close

- Provide documentation, deliver functional dishwashers, and offer training.
- Perform final system testing and User Acceptance Testing (UAT).
- Submit invoices, complete final budget and compare against initial.
- Evaluate project performance and document lessons learned.
- Store all project files in a central project management system.
- Acknowledge team contributions and reallocate resources.
- Conduct a closure meeting and obtain formal sign-off.